

Smart Academic Advisor System

Project Documentation

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1 Project Overview & Problem Statement

The **Smart Academic Advisor System** is an intelligent Desktop GUI application designed to help students plan their academic schedule efficiently. Acting as a virtual academic advisor, the system analyzes a student's current academic standing and completed coursework to recommend the optimal course load and schedule. The core objective is to prevent registration errors, enforce prerequisite rules, and tailor the credit load based on the student's GPA.

2 Development Methodology

The system was developed following a structured lifecycle:

- **Phase 1 & 2 (Requirements & Data Design):** Defined core inputs and structured course data (Name, Code, Prerequisites).
- **Phase 3 (Logic):** Built the Decision Engine to handle load determination.
- **Phase 4 & 5 (Flow & GUI):** Mapped the data through a JavaFX interface.
- **Phase 6 & 7 (QA & Snapshots):** Conducted rigorous testing and documented error cases.

3 System Inputs & User Profile

The application collects comprehensive data including:

- Personal Information (Name, ID).
- Academic Major (AI or CS) and specific tracks (Media, General, Big Data).
- Current Cumulative GPA and Academic Year.
- Dynamic checklist for previously passed courses.

4 Core Architecture: The Decision Engine

Rule 1: Load Determination

Low GPA (Half Load), Average GPA (Normal Load), High GPA (Overload).

Rule 2 & 3: Course Recommendation Logic

Courses are recommended based on prerequisites fulfillment. Delayed courses are given the highest priority to prevent graduation delays.

5 Error Checking & Exception Handling

- **Format Exceptions:** Validates GPA and ID inputs (restricting text).
- **Missing Inputs:** Checks for empty fields and displays warning alerts.

- **Logical Exceptions:** Bounds GPA between 0.0 and 4.0 and prevents contradictory states.

6 Technology Stack & Tools

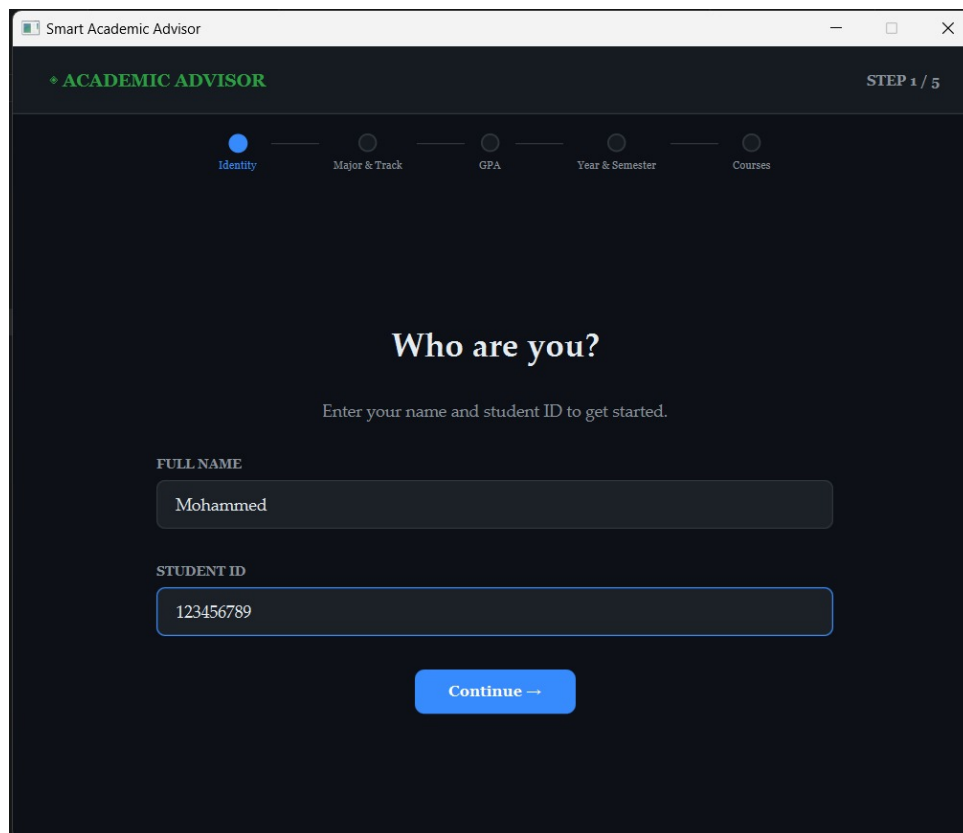
- **Programming Language:** Java (OOP)
- **GUI Framework:** JavaFX SDK 17
- **IDE:** Visual Studio Code
- **Data Storage:** JSON
- **Version Control:** Git & GitHub

7 Development Challenges & Issue Resolution

- **Data Parsing:** Resolved issues parsing complex course hierarchies in JSON.
- **Prerequisite Redundancy:** Refined backward-completion validation rules.
- **Track Overlap Logic:** Implemented strict conditional filtering to prevent Media subjects from appearing for Big Data students.
- **Environment Configuration:** Debugged JavaFX SDK 17 library path issues in VS Code.

8 System Snapshots (GUI)

This section demonstrates the user flow through the application.



The screenshot shows a web browser window titled "Smart Academic Advisor". The page has a dark blue background. At the top, there is a header with the text "ACADEMIC ADVISOR" in green and "STEP 1 / 5" in white. Below the header, there is a progress bar with five steps: "Identity", "Major & Track", "GPA", "Year & Semester", and "Courses". The "Identity" step is currently active, indicated by a blue dot. The main content area is titled "Who are you?" in white. Below this title, there is a subtitle "Enter your name and student ID to get started." in a smaller font. There are two input fields: "FULL NAME" with the value "Mohammed" and "STUDENT ID" with the value "123456789". Both fields have a light blue border. At the bottom of the form, there is a blue button labeled "Continue →".

Figure 1: System Landing Page

Smart Academic Advisor

ACADEMIC ADVISOR

STEP 2 / 5

Identity Major & Track GPA Year & Semester Courses

Your Major

Select your department and specialization track.

MAJOR

CS
 Computer Science

AI
 Artificial Intelligence

TRACK

BIGDATA
 Big Data

MEDIA
 Media

GENERAL
 General

← Back Continue →

Figure 2: Major & Track Selection (Computer Science)

Smart Academic Advisor

ACADEMIC ADVISOR

STEP 2 / 5

Identity Major & Track GPA Year & Semester Courses

Your Major

Select your department and specialization track.

MAJOR

CS
 Computer Science

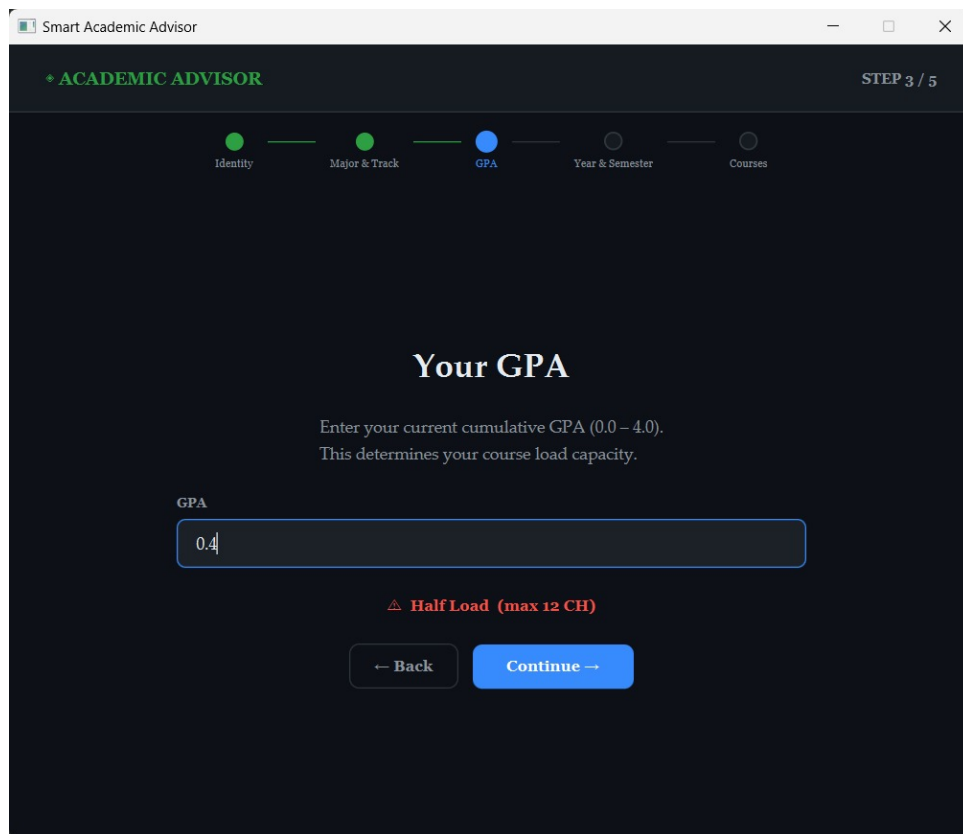
AI
 Artificial Intelligence

TRACK

AI
 AI Track

← Back Continue →

Figure 3: Major & Track Selection (Artificial Intelligence)



Smart Academic Advisor

ACADEMIC ADVISOR

STEP 3 / 5

Identity Major & Track GPA Year & Semester Courses

Your GPA

Enter your current cumulative GPA (0.0 – 4.0).
This determines your course load capacity.

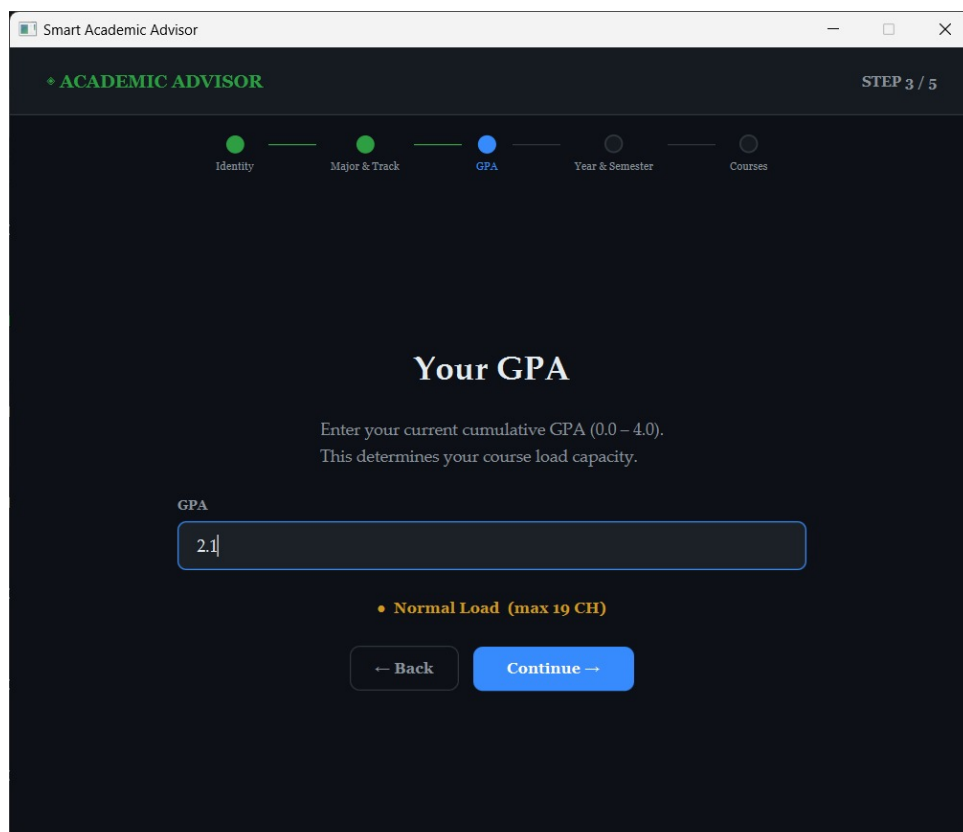
GPA

0.4

△ Half Load (max 12 CH)

← Back Continue →

Figure 4: GPA Entry & Academic Status (Half-Load Scenario)



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ACADEMIC ADVISOR

STEP 3 / 5

Identity Major & Track GPA Year & Semester Courses

Your GPA

Enter your current cumulative GPA (0.0 – 4.0).
This determines your course load capacity.

GPA

2.1

● Normal Load (max 19 CH)

← Back Continue →

Figure 5: GPA Entry & Academic Status (Normal-Load Scenario)

Smart Academic Advisor

ACADEMIC ADVISOR

STEP 3 / 5

Identity Major & Track GPA Year & Semester Courses

Your GPA

Enter your current cumulative GPA (0.0 – 4.0).
This determines your course load capacity.

GPA

3.4

★ Overload (max 21 CH)

← Back Continue →

Figure 6: GPA Entry & Academic Status (Overload Scenario)

Smart Academic Advisor

ACADEMIC ADVISOR

STEP 4 / 5

Identity Major & Track GPA Year & Semester Courses

Academic Standing

Select your current year and semester.

ACADEMIC YEAR

Year 1 1st Year

Year 2 2nd Year

Year 3 3rd Year

Year 4 4th Year

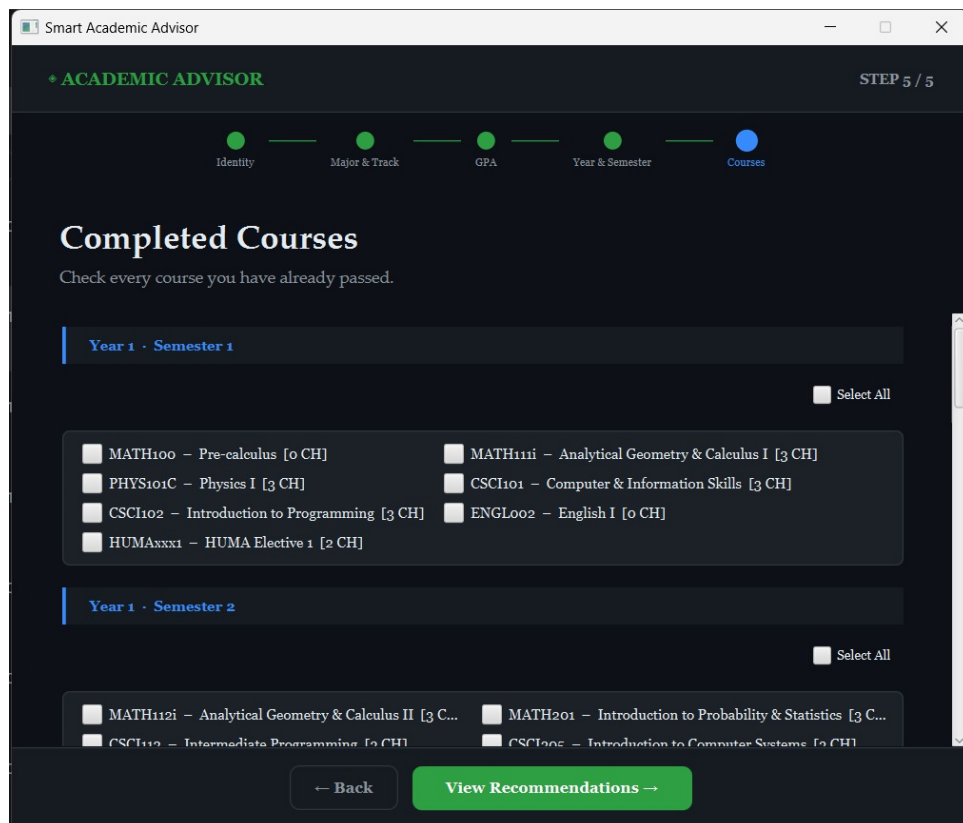
SEMESTER

Sem 1 1st Semester

Sem 2 2nd Semester

← Back Continue →

Figure 7: Year & Semester Selection



Smart Academic Advisor

ACADEMIC ADVISOR

STEP 5 / 5

Identity Major & Track GPA Year & Semester Courses

Completed Courses

Check every course you have already passed.

Year 1 - Semester 1

Select All

- ☐ MATH100 - Pre-calculus [0 CH]
- ☐ PHYS101C - Physics I [3 CH]
- ☐ CSCI102 - Introduction to Programming [3 CH]
- ☐ HUMAxxxx1 - HUMA Elective 1 [2 CH]
- ☐ MATH111i - Analytical Geometry & Calculus I [3 CH]
- ☐ CSCI101 - Computer & Information Skills [3 CH]
- ☐ ENGL002 - English I [0 CH]

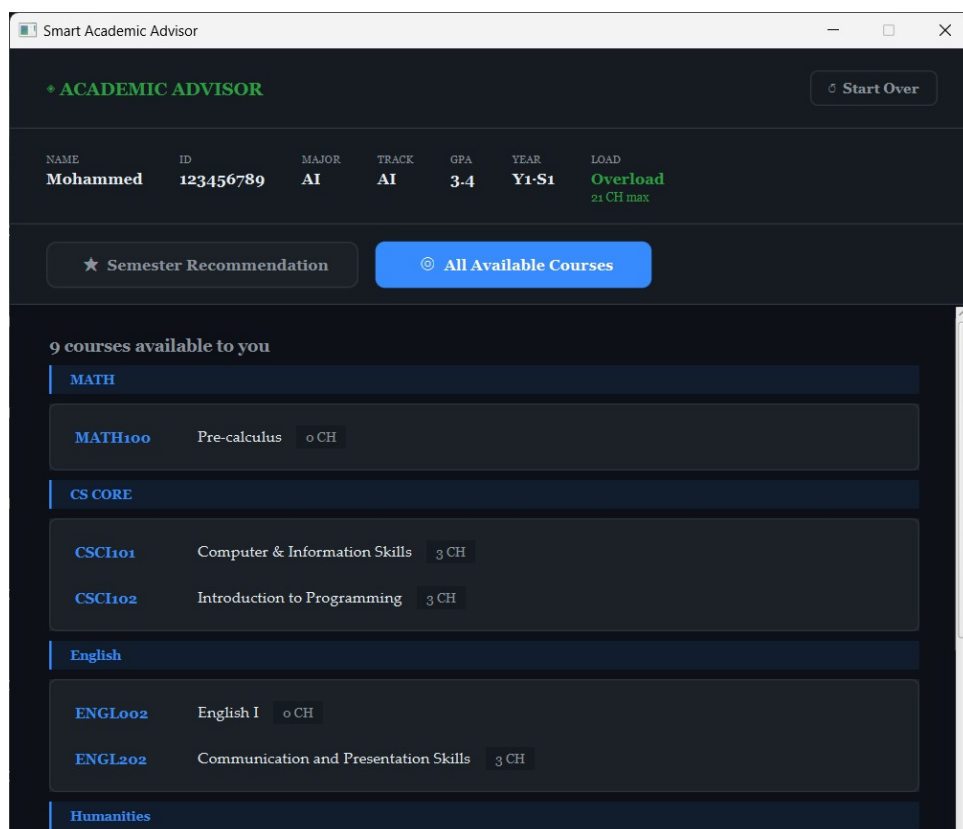
Year 1 - Semester 2

Select All

- ☐ MATH112i - Analytical Geometry & Calculus II [3 C...
- ☐ CSCI112 - Intermediate Programming [3 CH]
- ☐ MATH201 - Introduction to Probability & Statistics [3 C...
- ☐ CSCI205 - Introduction to Computer Systems [3 CH]

← Back View Recommendations →

Figure 8: Completed Courses Checklist



Smart Academic Advisor

ACADEMIC ADVISOR

Start Over

NAME	ID	MAJOR	TRACK	GPA	YEAR	LOAD
Mohammed	123456789	AI	AI	3-4	Y1-S1	Overload 21 CH max

★ Semester Recommendation All Available Courses

9 courses available to you

MATH

- MATH100 Pre-calculus 0 CH

CS CORE

- CSCI101 Computer & Information Skills 3 CH
- CSCI102 Introduction to Programming 3 CH

English

- ENGL002 English I 0 CH
- ENGL202 Communication and Presentation Skills 3 CH

Humanities

Figure 9: All Available Courses Directory

Smart Academic Advisor

ACADEMIC ADVISOR
Start Over

NAME	ID	MAJOR	TRACK	GPA	YEAR	LOAD
Mohammed	123456789	AI	AI	3.4	Y1-S1	Overload 21 CH max

★ Semester Recommendation

⦿ All Available Courses

FIXED

Recommended Courses

17 CH

CSCI102

Introduction to Programming

3 CH

MATH100

Pre-calculus

0 CH

CSCI101

Computer & Information Skills

3 CH

ENGL002

English I

0 CH

HUMAXXX1

HUMA Elective 1

2 CH

ENGL202

Communication and Presentation Skills

3 CH

HUMAXXX2

HUMA Elective 2

2 CH

SSCIXXX1

SSCI 1

2 CH

SSCIXXX2

SSCI 2

2 CH

Figure 10: Final Recommended Schedule (Decision Engine Output)

9 Project Repository & References

Source Code

The complete source code and project files are hosted on GitHub:

<https://github.com/tarek0026/Smart-Academic-Advisor-System>

Learning Resources & References

The following resources were utilized to study and implement the JavaFX GUI:

- JavaFX GUI Tutorial (Bro Code):
https://youtu.be/9XJicRt_FaI?si=Xz0AMETcgFqjcJuE
- JavaFX Playlist (Bucky / thenewboston):
https://youtube.com/playlist?list=PL6gx4Cwl9DGBzfXLWLSYVy8EbTdpGbUIG&si=d9r4IQ0Iwvt5XLr_
- Java Playlist (ProgrammingKnowledge):
<https://youtube.com/playlist?list=PL1DUmTEdeA6K7rdxKiWJq6JIxTvHalY8f&si=sqNrtCefcqMmlzeQ>